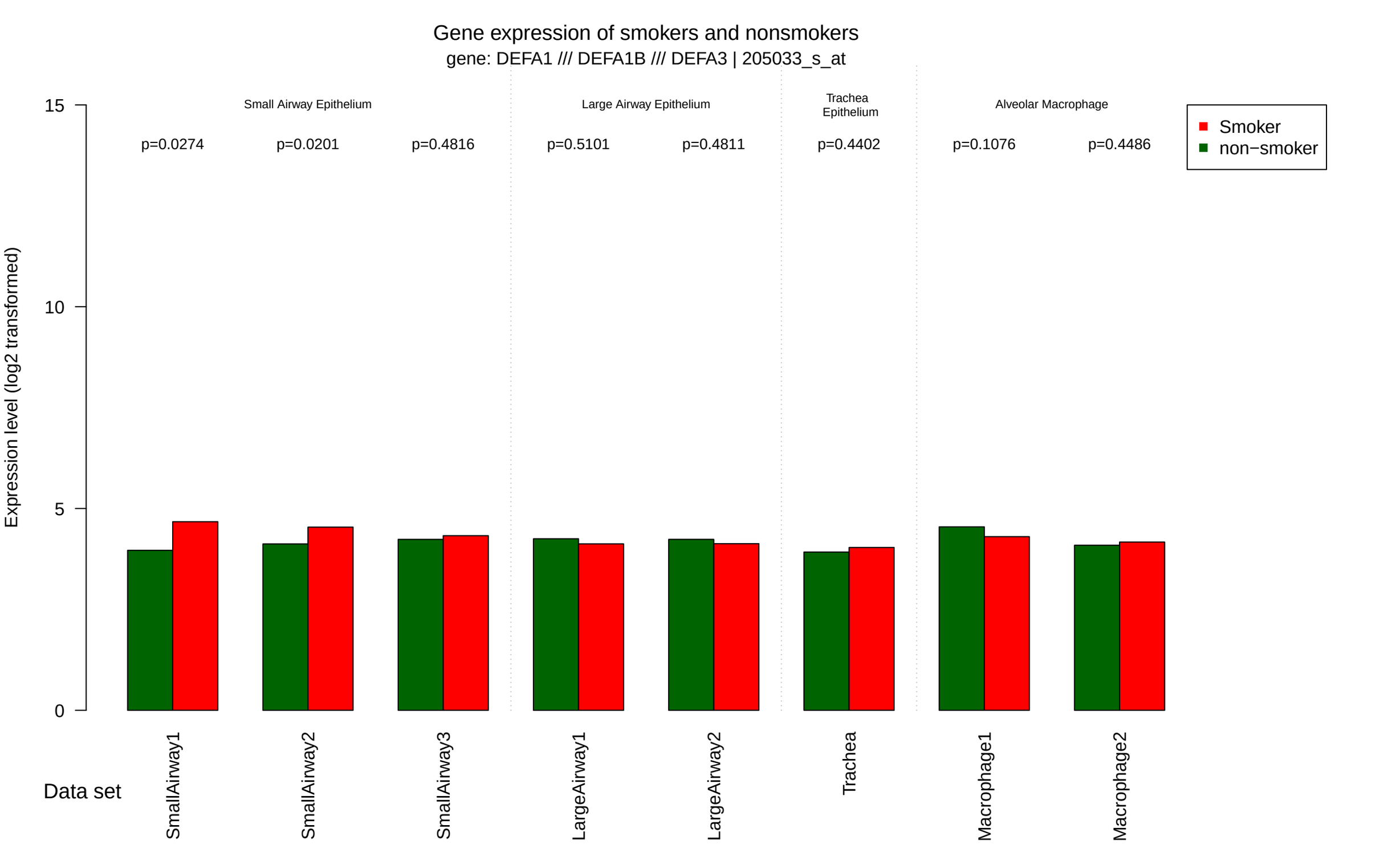
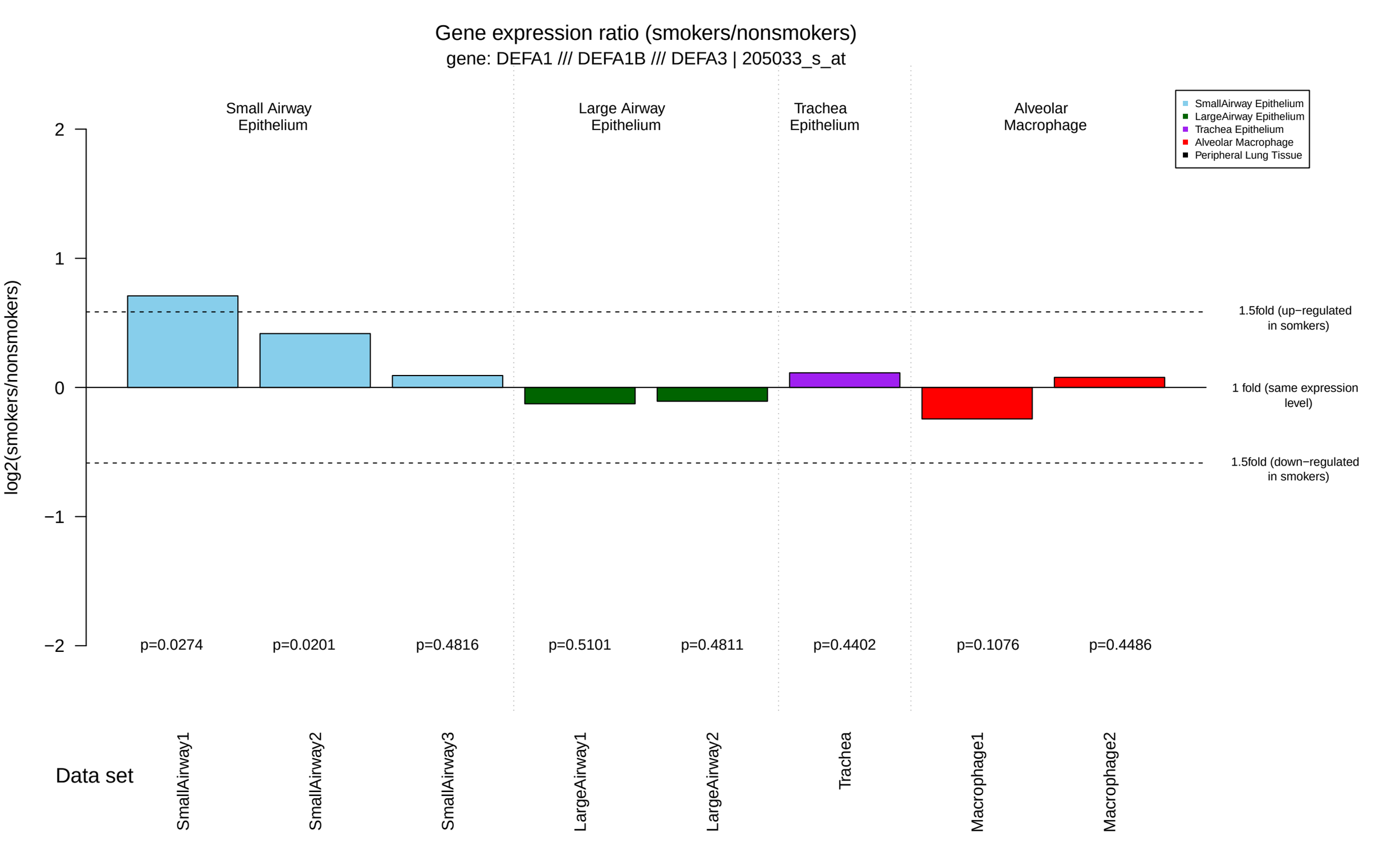
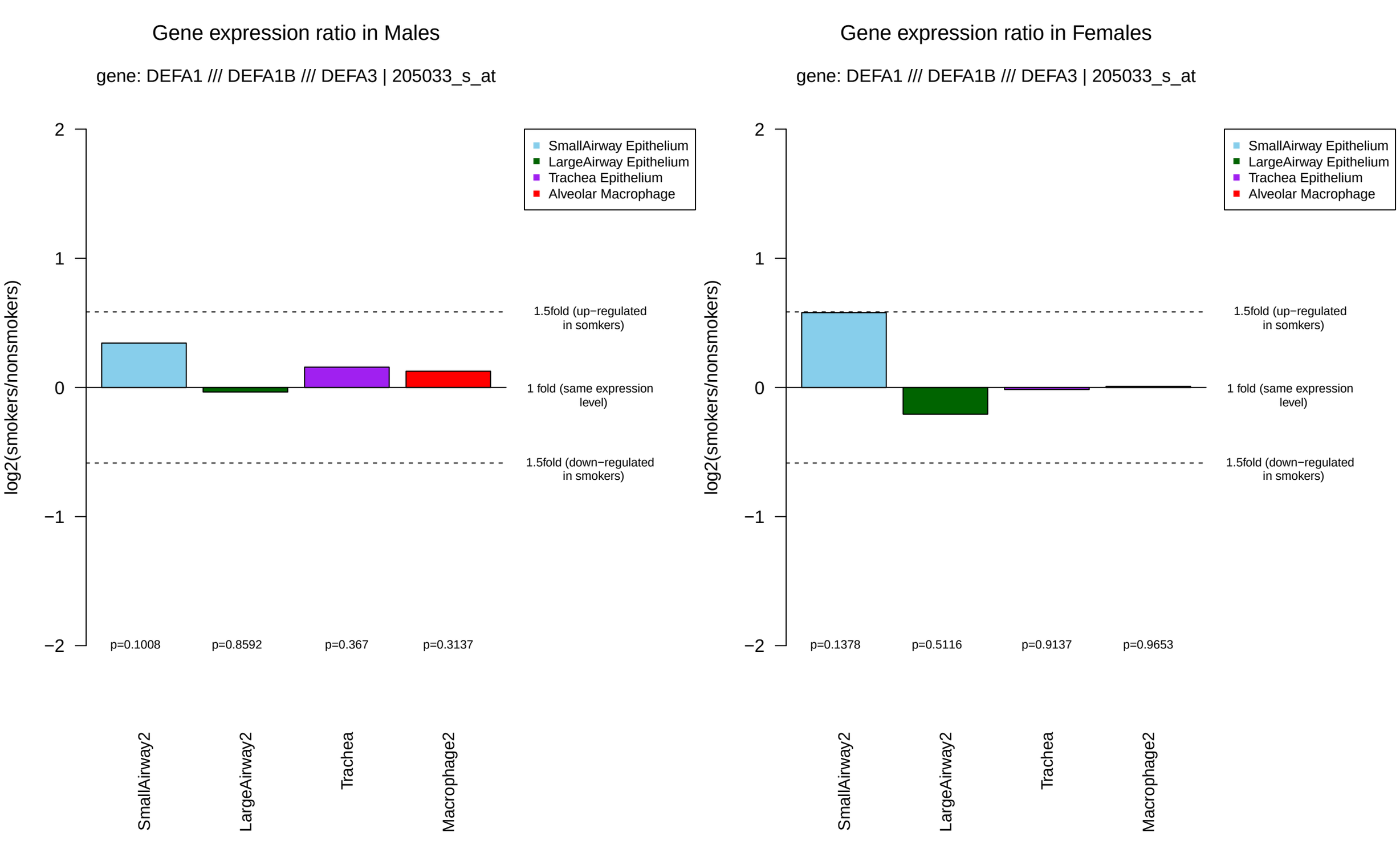
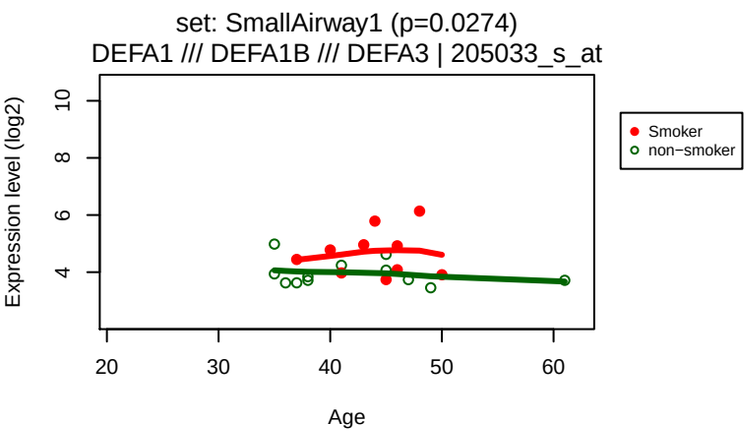
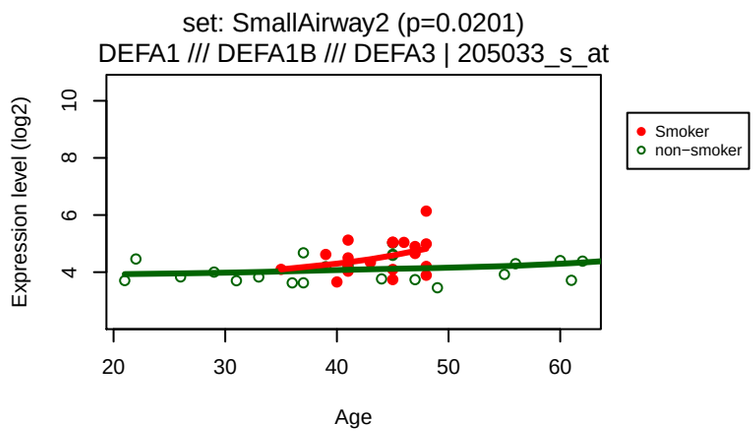
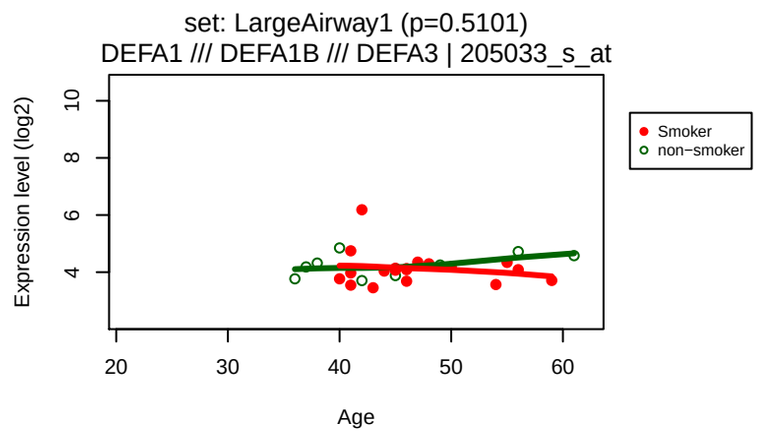

Gene expression of smokers and nonsmokers
gene: DEFA1 /// DEFA1B /// DEFA3 | 205033_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0274
p=0.0201
p=0.4816
p=0.5101
p=0.4811
p=0.4402
p=0.1076
p=0.4486
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

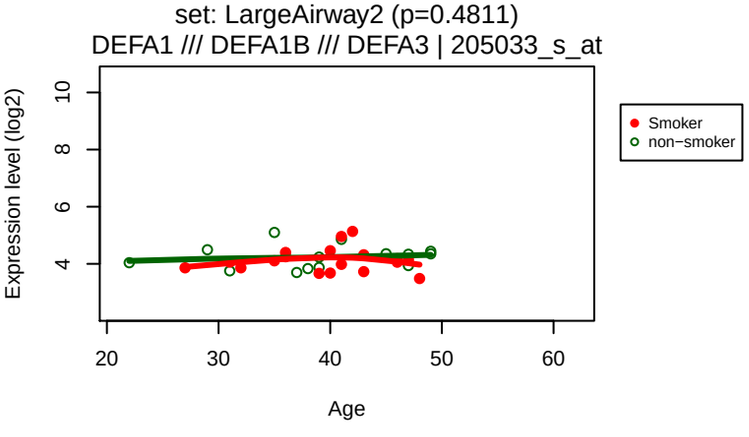

Gene expression ratio (smokers/nonsmokers)
gene: DEFA1 /// DEFA1B /// DEFA3 | 205033_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0274
p=0.0201
p=0.4816
p=0.5101
p=0.4811
p=0.4402
p=0.1076
p=0.4486
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

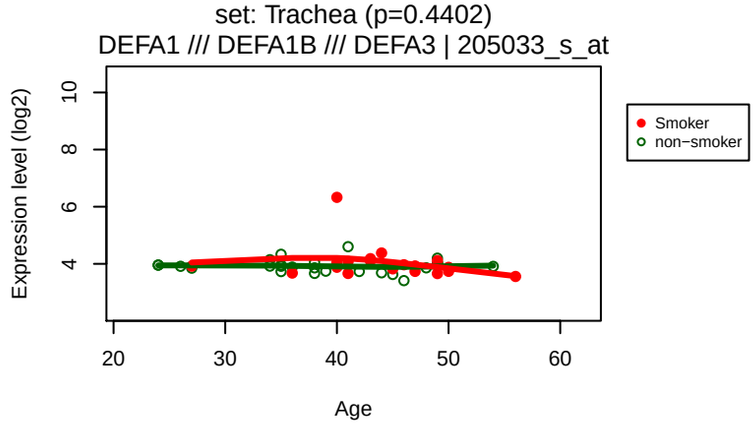

Gene expression ratio in Males
gene: DEFA1 /// DEFA1B /// DEFA3 | 205033_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1008
p=0.8592
p=0.367
p=0.3137
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DEFA1 /// DEFA1B /// DEFA3 | 205033_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1378
p=0.5116
p=0.9137
p=0.9653
SmallAirway2
LargeAirway2
Trachea
Macrophage2

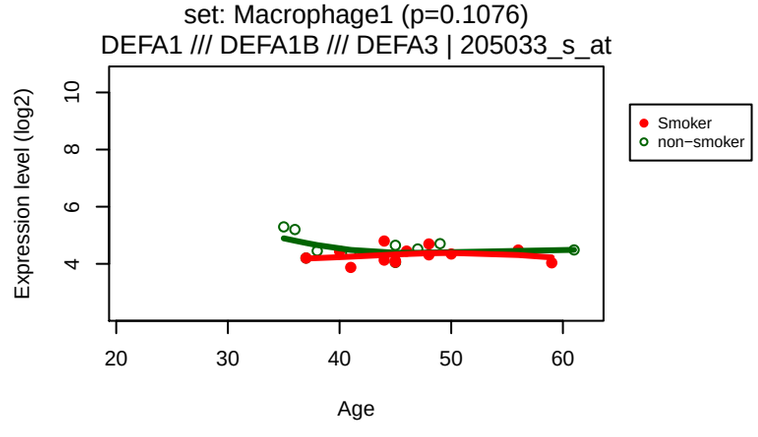

set: SmallAirway1 (p=0.0274)
DEFA1 /// DEFA1B /// DEFA3 | 205033_s_at
Expression level (log2)
Age
Smoker
non−smoker

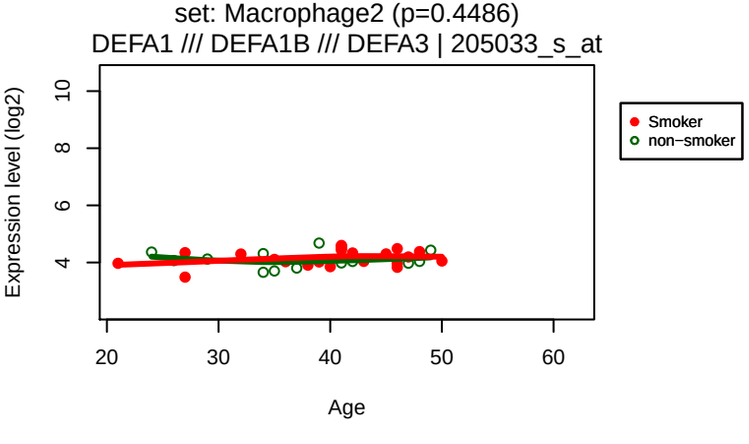

set: SmallAirway2 (p=0.0201)
DEFA1 /// DEFA1B /// DEFA3 | 205033_s_at
Expression level (log2)
Age
Smoker
non−smoker

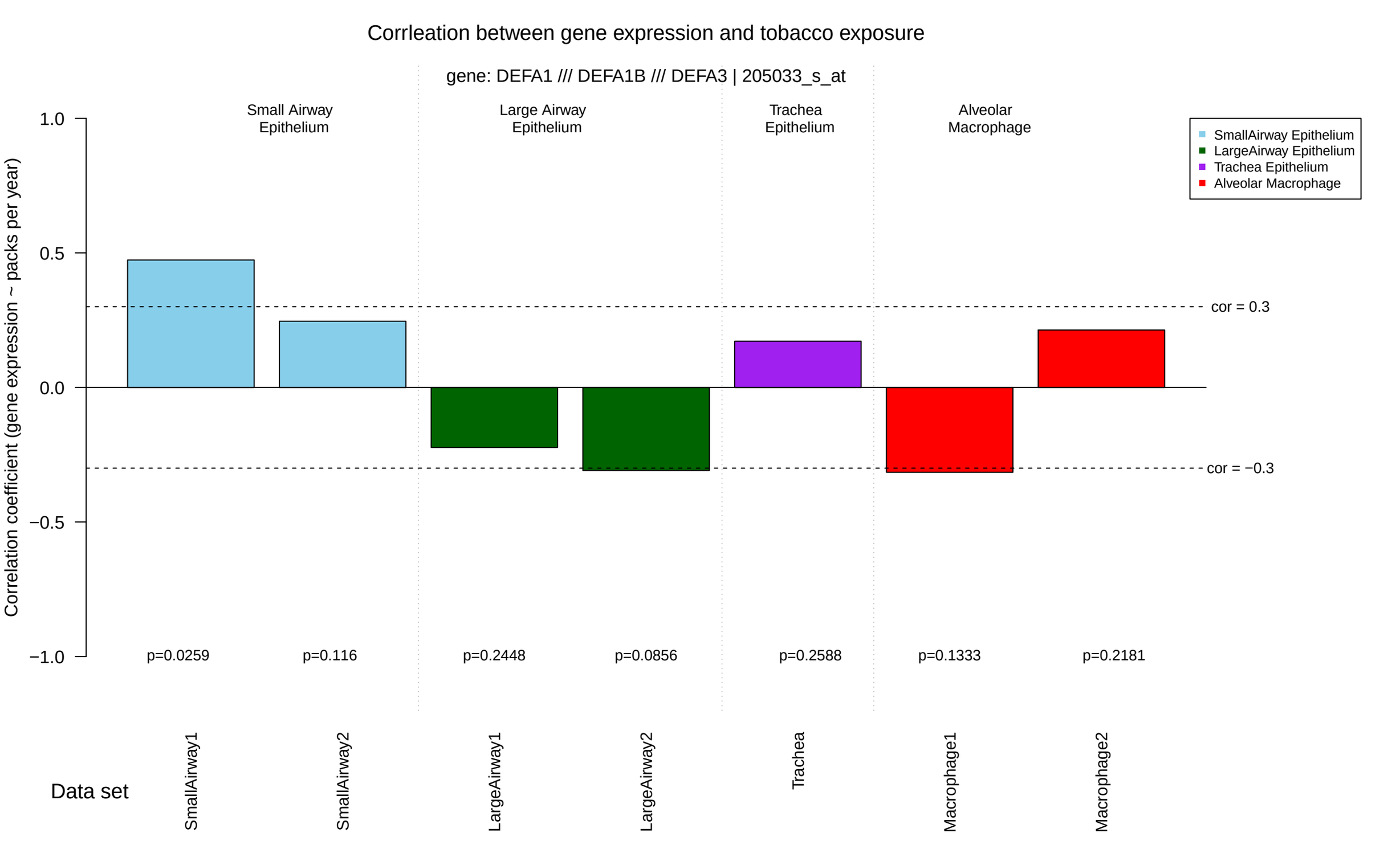

set: LargeAirway1 (p=0.5101)
DEFA1 /// DEFA1B /// DEFA3 | 205033_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4811)
DEFA1 /// DEFA1B /// DEFA3 | 205033_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4402)
DEFA1 /// DEFA1B /// DEFA3 | 205033_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1076)
DEFA1 /// DEFA1B /// DEFA3 | 205033_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4486)
DEFA1 /// DEFA1B /// DEFA3 | 205033_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DEFA1 /// DEFA1B /// DEFA3 | 205033_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0259
p=0.116
p=0.2448
p=0.0856
p=0.2588
p=0.1333
p=0.2181
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2